

Worksheet 1A: Fractions Fundamentals

Annotated Revision Guide

AIML Foundations Mathematics

Dublin and D'un Laoghaire ETB

Instructor: Josh Aaron

Part A: Numerical Fractions (Problems 1–12)

Concept. Problems 1–10 review the four operations on fractions. Problems 11–12 introduce **factorials**: $n! = n \times (n-1) \times \cdots \times 2 \times 1$. Since $n! = n \cdot (n-1)!$, consecutive factorials cancel gracefully: $\frac{n!}{(n-1)!} = n$.

Method. Add/subtract: find the LCD, convert, combine numerators. Multiply: cancel across numerator–denominator pairs before multiplying. Divide: multiply by the reciprocal. Mixed numbers: convert to improper fractions first.

Watch out. With mixed numbers, do *not* multiply whole and fractional parts separately: $2\frac{1}{3} \times 1\frac{1}{2} \neq 2 \times 1 + \frac{1}{3} \times \frac{1}{2}$. Convert first: $\frac{7}{3} \times \frac{3}{2} = \frac{7}{2}$.

Where this leads. Factorial fractions like $\frac{7!}{5! \cdot 2!}$ are binomial coefficients $\binom{7}{2} = 21$. These count combinations and are the coefficients in Pascal's Triangle — central to Worksheet 3C's probability section.

Explore further. Explore $\frac{(2n)!}{(n!)^2}$ — the central binomial coefficient. These grow like $4^n/\sqrt{\pi n}$ (Stirling's approximation). Also: why does $0! = 1$? Because the product of *no* numbers is 1 (the empty product).

1. $\frac{3}{4} + \frac{5}{6}$

LCD = 12. $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.

2. $\frac{7}{8} - \frac{2}{3}$

LCD = 24. $\frac{21}{24} - \frac{16}{24} = \frac{5}{24}$.

3. $\frac{5}{9} \times \frac{3}{10}$

Cancel first. $\frac{5^1}{9} \times \frac{3^1}{10^2} = \frac{1}{6}$.

Note: 5 cancels with 10; 3 cancels with 9.

4. $\frac{4}{5} \div \frac{2}{15}$

Invert and multiply. $\frac{4}{5} \times \frac{15}{2} = \frac{\cancel{4}^2}{\cancel{5}_1} \times \frac{\cancel{15}^3}{2^1} = 6.$

5. $\frac{2}{3} + \frac{1}{4} - \frac{1}{6}$

LCD = 12. $\frac{8}{12} + \frac{3}{12} - \frac{2}{12} = \frac{9}{12} = \frac{3}{4}.$

6. $\frac{7}{12} \times \frac{8}{21} \times \frac{3}{2}$

Cancel across all three. $\frac{7^1}{\cancel{12}^1} \times \frac{8^2}{\cancel{21}^3} \times \frac{3^1}{2^1} = \frac{2}{3}.$

Note: Cancel 7 with 21, 8 with 12 (leaving 2 and 1 respectively in those slots, but recheck carefully: $7/12 \times 8/21 \times 3/2$: 7 and 21 give $1/3$; 8 and 12 give $2/3$; 3 and 2 give $3/2$. Product: $\frac{1}{3} \cdot \frac{2}{3} \cdot \frac{3}{2} = \frac{6}{18} = \frac{1}{3} \cdot 2 = \frac{2}{3}.$

7. $\frac{\frac{3}{4}}{\frac{9}{16}}$

Divide. $\frac{3}{4} \div \frac{9}{16} = \frac{3}{4} \times \frac{16}{9} = \frac{\cancel{3}^1}{4^1} \times \frac{\cancel{16}^4}{9^3} = \frac{4}{3}.$

8. $\frac{\frac{2}{5} + \frac{1}{3}}{\frac{1}{2}}$

Numerator. $\frac{2}{5} + \frac{1}{3} = \frac{6}{15} + \frac{5}{15} = \frac{11}{15}.$

Divide. $\frac{11}{15} \div \frac{1}{2} = \frac{11}{15} \times 2 = \frac{22}{15}.$

9. $2\frac{1}{3} \times 1\frac{1}{2}$

Convert. $\frac{7}{3} \times \frac{3}{2} = \frac{7 \cdot \cancel{3}}{\cancel{3} \cdot 2} = \frac{7}{2} = 3\frac{1}{2}.$

10. $4\frac{2}{5} \div 1\frac{1}{10}$

Convert. $\frac{22}{5} \div \frac{11}{10} = \frac{22}{5} \times \frac{10}{11} = \frac{\cancel{22}^2}{\cancel{5}_1} \times \frac{\cancel{10}^2}{\cancel{11}^1} = 4.$

11. $\frac{5!}{4!}$

Cancel. $\frac{5 \times \cancel{4!}}{\cancel{4!}} = 5.$

Note: $n!/(n-1)! = n$ always.

12. $\frac{7!}{5! \cdot 2!}$

Cancel. $\frac{7 \times 6 \times \cancel{5!}}{\cancel{5!} \times 2} = \frac{42}{2} = 21.$

Note: This is $\binom{7}{2} = 21$ — the number of ways to choose 2 items from 7.

Part B: Pattern Sums (Problems 13–20)

Concept. Two powerful ideas: the **geometric series** $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ (for $|r| < 1$), and the **telescoping sum**, where adjacent terms cancel. Problems 13–16 are geometric with ratio $\frac{1}{2}$; problems 17–19 telescope; problem 20 builds the partial sum of $e = \sum_{n=0}^{\infty} \frac{1}{n!}$.

Method. Geometric partial sum: $S_n = \frac{1-r^{n+1}}{1-r}$. Telescoping: write $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, then most terms cancel. Factorial series: compute each term $\frac{1}{k!}$ and add over a common denominator.

Watch out. $1 + \frac{1}{2} + \frac{1}{4} + \cdots$ starts at the term $1 = \left(\frac{1}{2}\right)^0$. The formula $\frac{1}{1-r} = 2$ gives the limit as the number of terms *grows without bound* — it is not a finite calculation.

Where this leads. The e -series in problem 20 is the Taylor expansion of e^x at $x = 1$. Worksheet 3C explores Taylor polynomials for $\sin$, $\cos$, and e^x : these are just truncated infinite series.

Explore further. Read about *Basel problem*: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, proved by Euler in 1735. The telescoping technique also underpins partial fraction decomposition in calculus integration.

13. $1 + \frac{1}{2} + \frac{1}{4}$

LCD = 4. $\frac{4}{4} + \frac{2}{4} + \frac{1}{4} = \frac{7}{4} = 1.75.$

Note: Equivalently, $S_2 = \frac{1-(1/2)^3}{1-1/2} = \frac{7/8}{1/2} = \frac{7}{4}.$

14. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

LCD = 8. $\frac{8+4+2+1}{8} = \frac{15}{8} = 1.875.$

Look ahead: The pattern of numerators: 3, 7, 15, 31 ... These are $2^n - 1$.

15. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$

$$\text{LCD} = 16. \quad \frac{16 + 8 + 4 + 2 + 1}{16} = \frac{31}{16} = 1.9375.$$

16. What value does $1 + \frac{1}{2} + \frac{1}{4} + \cdots$ approach?

Geometric series. Ratio $r = \frac{1}{2}$, first term $a = 1$.

Formula. $S = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2.$

Note: The partial sums 1.75, 1.875, 1.9375 ... are creeping toward 2 but never quite arriving.

17. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4}$

Partial fractions. $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$

Telescope. $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) = 1 - \frac{1}{4} = \frac{3}{4}.$

Note: Only the first and last terms survive: $1 - \frac{1}{4}$.

18. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5}$

Telescope. $1 - \frac{1}{5} = \frac{4}{5}.$

Look ahead: The sum of n terms is always $1 - \frac{1}{n+1} = \frac{n}{n+1}.$

19. If we add terms $\frac{1}{n(n+1)}$ forever, what is the limit?

Pattern. $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \rightarrow 1.$

Formally. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N+1}\right) = 1.$

20. $1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!}$

Evaluate each term. $1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}.$

LCD = 24. $\frac{24 + 24 + 12 + 4 + 1}{24} = \frac{65}{24} \approx 2.708.$

Note: The full infinite series $\sum_{n=0}^{\infty} \frac{1}{n!} = e \approx 2.71828 \dots$. Adding just 5 terms already gives 97% accuracy.

Part C: Algebraic Fractions — Simplification (Problems 21–30)

Concept. An algebraic fraction simplifies when numerator and denominator share a *polynomial* common factor. The process: (1) factor numerator and denominator completely; (2) cancel identical factors. Factorial expressions simplify by writing $n! = n \cdot (n-1)!$ etc.

Method. For monomials: cancel common powers of each variable. For polynomials: factor quadratics using the standard methods (difference of squares, product–sum, grouping). For factorials: expand only as far as needed to see the cancellation.

Watch out. $\frac{x^2 - 4}{x + 2}$: factor the numerator first. Never cancel terms that are added together: $\frac{x^2 + 4}{x + 2} \neq x + 2$. Only *factors* (things multiplied) can cancel.

Where this leads. Simplifying algebraic fractions is the gateway to rational functions in calculus — finding limits, asymptotes, and integration by partial fractions. The factorial fractions here are binomial coefficients $\binom{n}{k}$, essential for the probability content in Worksheet 3C.

Explore further. The Fundamental Theorem of Algebra guarantees every polynomial over $\mathbb{C}$ factors completely into linear factors. Over $\mathbb{R}$, irreducible quadratics (negative discriminant) cannot be factored further.

21. $\frac{6p^2}{9p}$

Cancel. $\frac{\cancel{3} \cdot 2 \cdot p \cdot \cancel{p}}{\cancel{3} \cdot 3 \cdot \cancel{p}} = \frac{2p}{3}.$

22. $\frac{15m^3n^2}{25mn^4}$

Cancel. $\frac{\cancel{5} \cdot 3 \cdot m^{\cancel{3}^2} \cdot \cancel{n^2}}{\cancel{5} \cdot 5 \cdot \cancel{m} \cdot \cancel{n^2} \cdot n^2} = \frac{3m^2}{5n^2}.$

23. $\frac{t^2 - 4}{t + 2}$

Factor. $t^2 - 4 = (t - 2)(t + 2).$

Cancel. $\frac{(t - 2)\cancel{(t + 2)}}{\cancel{(t + 2)}} = t - 2.$

24. $\frac{r^2 - 9}{r^2 + 6r + 9}$

Factor. $(r - 3)(r + 3)$ over $(r + 3)^2.$

Cancel. $\frac{(r - 3)\cancel{(r + 3)}}{\cancel{(r + 3)}(r + 3)} = \frac{r - 3}{r + 3}.$

25. $\frac{2k^2 + 4k}{k^2 + 2k}$

Factor. $\frac{2k(k+2)}{k(k+2)}$.

Cancel. $\frac{\cancel{2k(k+2)}}{\cancel{k(k+2)}} = 2$.

26. $\frac{h^2 - 5h + 6}{h^2 - 4}$

Factor. $(h-2)(h-3)$ over $(h-2)(h+2)$.

Cancel. $\frac{\cancel{(h-2)}(h-3)}{\cancel{(h-2)}(h+2)} = \frac{h-3}{h+2}$.

27. $\frac{n!}{(n-1)!}$

Expand one step. $\frac{n \cdot \cancel{(n-1)!}}{\cancel{(n-1)!}} = n$.

Note: This is why $n! = n \cdot (n-1)!$ — the key recurrence.

28. $\frac{(n+1)!}{n!}$

Same idea. $\frac{(n+1) \cdot \cancel{n!}}{\cancel{n!}} = n+1$.

29. $\frac{(n+2)!}{n! \cdot 2!}$

Expand. $\frac{(n+2)(n+1)\cancel{n!}}{\cancel{n!} \cdot 2} = \frac{(n+2)(n+1)}{2}$.

Note: This is $\binom{n+2}{2}$, the number of ways to choose 2 items from $n+2$.

30. $\frac{6s^2 - s - 2}{2s^2 + 3s - 2}$

Factor numerator. $6s^2 - s - 2 = (3s-2)(2s+1)$.

Factor denominator. The answer key intends denominator $2s^2 + 5s + 2 = (2s+1)(s+2)$.

Cancel. $\frac{(3s-2)\cancel{(2s+1)}}{\cancel{(2s+1)}(s+2)} = \frac{3s-2}{s+2}$.

*Note: **Note on source:** As printed, the denominator $2s^2 + 3s - 2 = (2s-1)(s+2)$, which shares no factor with the numerator. The answer key answer $\frac{3s-2}{s+2}$ requires denominator*

$2s^2 + 5s + 2$; this appears to be a typo in the worksheet. The intended lesson is illustrated above.

Part D: Algebraic Fractions — Operations (Problems 31–40)

Concept. Adding/subtracting algebraic fractions follows the same logic as numerical fractions: convert to a common denominator, then combine numerators. Multiplying and dividing: factor first, cancel across numerator–denominator pairs, then multiply what remains.

Method. Find LCD as the product of all distinct factors (each to its highest power). For $\frac{A}{B} \div \frac{C}{D}$: rewrite as $\frac{A}{B} \times \frac{D}{C}$. For complex chains (problem 39): work left to right, converting each division to multiplication before proceeding.

Watch out. When subtracting, distribute the minus sign across the whole numerator: $\frac{5}{t+1} - \frac{2}{t-1} = \frac{5(t-1) - 2(t+1)}{(t+1)(t-1)}$; do not write $5(t-1) + 2(t+1)$ in the numerator by mistake.

Where this leads. Problem 40 — solving $\frac{1}{m} + \frac{1}{n} = \frac{1}{k}$ — gives the formula for parallel resistors, combined focal lengths, and the harmonic mean. Clearing fractions to solve *fractional equations* is a key calculus tool.

Explore further. Rational expressions are the analogue of rational numbers for polynomials. The set of all rational functions $\mathbb{R}(x)$ forms a *field* — it is closed under $+$, $-$, $\times$, $\div$ (excluding division by zero). See Stewart, *Precalculus*, Ch. P.6 for further examples.

31. $\frac{2}{p} + \frac{3}{q}$

$$\text{LCD} = pq. \quad \frac{2q}{pq} + \frac{3p}{pq} = \frac{2q + 3p}{pq}.$$

32. $\frac{5}{t+1} - \frac{2}{t-1}$

$$\text{LCD} = (t+1)(t-1). \quad \frac{5(t-1) - 2(t+1)}{(t+1)(t-1)} = \frac{5t - 5 - 2t - 2}{t^2 - 1} = \frac{3t - 7}{t^2 - 1}.$$

33. $\frac{3}{m} + \frac{2}{n} - \frac{1}{mn}$

$$\text{LCD} = mn. \quad \frac{3n + 2m - 1}{mn}.$$

34. $\frac{r}{r+2} + \frac{2}{r-2}$

$$\text{LCD} = (r+2)(r-2). \quad \frac{r(r-2) + 2(r+2)}{(r+2)(r-2)} = \frac{r^2 - 2r + 2r + 4}{r^2 - 4} = \frac{r^2 + 4}{(r+2)(r-2)}.$$

35. $\frac{1}{k-1} - \frac{1}{k+1}$

LCD $= (k-1)(k+1) = k^2 - 1$. $\frac{(k+1) - (k-1)}{k^2 - 1} = \frac{2}{k^2 - 1}$.

36. $\frac{2s}{s+3} \times \frac{s^2-9}{4s^2}$

Factor. $s^2 - 9 = (s-3)(s+3)$.

Cancel. $\frac{\cancel{2s}}{\cancel{(s+3)}} \times \frac{(s-3)\cancel{(s+3)}}{4\cancel{s} \cdot s} = \frac{s-3}{2s}$.

37. $\frac{w^2-4}{w+5} \div \frac{w-2}{w^2+5w}$

Factor. $w^2 - 4 = (w-2)(w+2)$; $w^2 + 5w = w(w+5)$.

Rewrite. $\frac{(w-2)(w+2)}{\cancel{(w+5)}} \times \frac{\cancel{w(w+5)}}{\cancel{(w-2)}} = w(w+2) = w^2 + 2w$.

38. $\frac{1}{u} + \frac{1}{u+1} + \frac{1}{u+2}$

LCD $= u(u+1)(u+2)$. $\frac{(u+1)(u+2) + u(u+2) + u(u+1)}{u(u+1)(u+2)}$.

Expand numerator. $(u^2 + 3u + 2) + (u^2 + 2u) + (u^2 + u) = 3u^2 + 6u + 2$.

Result. $\frac{3u^2 + 6u + 2}{u(u+1)(u+2)}$.

39. $\frac{p^2-1}{p^2+2p} \cdot \frac{p^2+4p+4}{p^2+p-2} \div \frac{p+2}{p}$

Factor all. $p^2 - 1 = (p-1)(p+1)$; $p^2 + 2p = p(p+2)$; $p^2 + 4p + 4 = (p+2)^2$; $p^2 + p - 2 = (p+2)(p-1)$.

Div to mult. $\frac{(p-1)(p+1)}{p(p+2)} \cdot \frac{(p+2)^2}{(p+2)(p-1)} \cdot \frac{p}{(p+2)}$.

Cancel. $\frac{\cancel{(p-1)}(p+1)}{\cancel{p}(p+2)} \cdot \frac{\cancel{(p+2)}^2}{\cancel{(p+2)}(p-1)} \cdot \frac{\cancel{p}}{\cancel{(p+2)}} = \frac{p+1}{p+2}$.

40. If $\frac{1}{m} + \frac{1}{n} = \frac{1}{k}$, solve for k .

Add LHS. $\frac{m+n}{mn} = \frac{1}{k}$.

Reciprocal. $k = \frac{mn}{m+n}$.

Note: This is the harmonic combination formula: resistors in parallel, lenses in series.

Part E: Complex Fractions (Problems 41–46)

Concept. A **complex fraction** has a fraction in its numerator, denominator, or both. The fastest method: identify the LCD of all *inner* fractions, then multiply the whole expression (top and bottom) by that LCD. This clears all inner fractions in one step.

Method. (1) Identify every inner fraction. (2) Find the LCD of all their denominators. (3) Multiply numerator and denominator of the outer fraction by the LCD. (4) Simplify what remains.

Watch out. Problem 45 has a fraction nested three levels deep. Work from the *innermost* fraction outward. Never attempt to cancel across different levels before simplifying each level.

Where this leads. The continued fraction in problem 45, $\frac{1}{1+\frac{1}{1+\frac{1}{k}}}$, is the first step of a *continued fraction* expansion. Infinite continued fractions represent irrational numbers like $\sqrt{2} = [1; 2, 2, 2, \dots]$ and the golden ratio $\phi = [1; 1, 1, 1, \dots]$.

Explore further. Investigate: what is the value of $x = \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$? Setting

$x = \frac{1}{1+x}$ gives $x^2 + x - 1 = 0$, so $x = \frac{\sqrt{5}-1}{2} = \phi - 1$.

41.
$$\frac{\frac{1}{p} + \frac{1}{q}}{\frac{1}{p} - \frac{1}{q}}$$

Multiply by pq . $\frac{q+p}{q-p}$.

42.
$$\frac{\frac{2}{t+1} - \frac{1}{t}}{\frac{3}{t}}$$

Numerator (LCD= $t(t+1)$). $\frac{2t - (t+1)}{t(t+1)} = \frac{t-1}{t(t+1)}$.

Divide by $\frac{3}{t}$. $\frac{t-1}{t(t+1)} \times \frac{t}{3} = \frac{t-1}{3(t+1)}$.

43.
$$\frac{1 - \frac{1}{n^2}}{1 + \frac{1}{n}}$$

Multiply by n^2 . $\frac{n^2 - 1}{n^2 + n} = \frac{(n-1)\cancel{(n+1)}}{n\cancel{(n+1)}} = \frac{n-1}{n}$.

44.
$$\frac{\frac{m}{n} + \frac{n}{m}}{\frac{m^2 - n^2}{mn}}$$

Numerator (LCD= mn). $\frac{m^2 + n^2}{mn}$.

Divide. $\frac{m^2 + n^2}{mn} \times \frac{mn}{m^2 - n^2} = \frac{m^2 + n^2}{m^2 - n^2}$.

45. $\frac{1}{1 + \frac{1}{1 + \frac{1}{k}}}$

Innermost. $1 + \frac{1}{k} = \frac{k+1}{k}$.

Middle layer. $1 + \frac{k}{k+1} = \frac{2k+1}{k+1}$.

Outer. $\frac{1}{\frac{2k+1}{k+1}} = \frac{k+1}{2k+1}$.

46. $\frac{\frac{1}{h} - \frac{1}{h+1}}{h}$

Numerator. $\frac{(h+1) - h}{h(h+1)} = \frac{1}{h(h+1)}$.

Divide by h . $\frac{1}{h(h+1)} \div h = \frac{1}{h^2(h+1)}$.

*Note: **Correction:** The source answer key gives $\frac{1}{h(h+1)}$, which would be the numerator before dividing by h . The correct result after dividing is $\frac{1}{h^2(h+1)}$.*

Part F: Repeated Multiplication — Shrinking vs Growing (Problems 47–52)

Concept. When you repeatedly multiply a fraction r : if $|r| < 1$ (fraction less than 1), powers $r^n \rightarrow 0$; if $|r| > 1$ (fraction greater than 1), powers $r^n \rightarrow \infty$. The boundary case $|r| = 1$ stays constant. This is the heart of **geometric sequences and series**.

Method. $(r)^n$: multiply r by itself n times. For the sum $r + r^2 + \dots + r^n = r \cdot \frac{1-r^n}{1-r}$. As $n \rightarrow \infty$: sum $\rightarrow \frac{r}{1-r}$ if $|r| < 1$.

Watch out. $\left(\frac{99}{100}\right)^{100}$ looks close to 1 since $\frac{99}{100}$ is close to 1. But repeated small reductions compound dramatically: the result is $\approx 0.366 \approx e^{-1}$, far from 1. This counterintuitive behaviour is why compound interest and exponential decay matter so much.

Where this leads. $(1 - \frac{1}{n})^n \rightarrow e^{-1}$ as $n \rightarrow \infty$ (compare Worksheet 1B problem 17). Exponential decay in physics, finance, and ML all reduce to r^n with $|r| < 1$.

Explore further. The fact that $(1 + \frac{1}{n})^n \rightarrow e$ and $(1 - \frac{1}{n})^n \rightarrow e^{-1}$ can be proved using the definition of e as $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$. See Spivak, *Calculus*, Ch. 18.

47. Powers of $\frac{3}{4}$: (a) $\frac{3}{4}$ (b) $\left(\frac{3}{4}\right)^2$ (c) $\left(\frac{3}{4}\right)^3$ (d) $\left(\frac{3}{4}\right)^4$ (e) limit as $n \rightarrow \infty$.

Values. (a) 0.75 (b) 0.5625 (c) ≈ 0.4219 (d) ≈ 0.3164

Limit. (e) Since $\frac{3}{4} < 1$, $\left(\frac{3}{4}\right)^n \rightarrow 0$.

Note: Each step multiplies by $\frac{3}{4}$, shrinking the result by 25%.

48. Powers of $\frac{5}{4}$: (a)–(d) as above, (e) limit.

Values. (a) 1.25 (b) 1.5625 (c) ≈ 1.9531 (d) ≈ 2.4414

Limit. (e) Since $\frac{5}{4} > 1$, $\left(\frac{5}{4}\right)^n \rightarrow \infty$.

49. Complete the rule about repeated multiplication of fractions.

Rule. Less than 1 $\Rightarrow$ result approaches **0**. Greater than 1 $\Rightarrow$ result grows toward ∞ .

50. $\left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4$

Values. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{8+4+2+1}{16} = \frac{15}{16} = 0.9375$.

Look ahead: The full infinite sum $\sum_{n=1}^{\infty} (1/2)^n = 1$. Adding four terms already reaches 93.75%.

51. Estimate $\left(\frac{9}{10}\right)^{10}$.

Reasoning. $\frac{9}{10} = 0.9$; ten multiplications of 0.9.

Calculation. $0.9^{10} \approx 0.3487$. Closer to 0.5, but below it.

Note: This is $(1 - \frac{1}{10})^{10}$, a step toward $e^{-1} \approx 0.368$.

52. Estimate $\left(\frac{99}{100}\right)^{100}$ without full calculation.

Key insight. $(1 - \frac{1}{n})^n \rightarrow e^{-1} \approx 0.368$ as $n \rightarrow \infty$.

Estimate. With $n = 100$: ≈ 0.366 . Much closer to $\frac{1}{3}$ than to 1.

Note: Even though each step reduces by only 1%, one hundred such steps compound to a 63% reduction. This is the mathematics behind radioactive decay, loan amortisation, and dropout regularisation in neural networks.

Part G: Limiting Behaviour (Problems 53–60)

Concept. These problems develop **limit intuition** — how fractions behave as a variable grows very large or approaches zero. Three key patterns: (1) $\frac{\text{constant}}{n} \rightarrow 0$

as $n \rightarrow \infty$; (2) $\frac{n}{\text{constant}} \rightarrow \infty$ as $n \rightarrow \infty$; (3) dominant terms: $\frac{n^2}{n^2+1} \approx \frac{n^2}{n^2} = 1$ for large n .

Method. For large n : identify the *dominant* (fastest-growing) term in numerator and denominator. For $\frac{n^2}{n^2+1}$, divide top and bottom by n^2 to get $\frac{1}{1+1/n^2} \rightarrow 1$. For small n : substitute small values and watch the behaviour.

Watch out. $\frac{n}{n+1}$ never equals 1, even for huge n . It *approaches* 1 as a limit. The distinction between "reaching" and "approaching" a value is the essence of the limit concept in calculus.

Where this leads. These are informal limits. Calculus formalises them with ε - δ definitions. The behaviours here — horizontal asymptotes, vertical asymptotes, limits at zero — are exactly what you will compute with L'Hôpital's Rule and Taylor series.

Explore further. The sigmoid and softmax functions in Worksheet 1B are built entirely from these limit patterns. A neural network's activation functions are designed to have convenient limiting behaviour (output bounded between 0 and 1, or summing to 1).

53. $\frac{n}{100}$ for $n > 0$: (a) as $n \rightarrow \infty$; (b) as $n \rightarrow 0^+$.

(a) Grows without bound: $\frac{n}{100} \rightarrow \infty$.

(b) Approaches 0: $\frac{n}{100} \rightarrow 0$.

54. $\frac{100}{n}$ for $n > 0$: (a) as $n \rightarrow \infty$; (b) as $n \rightarrow 0^+$.

(a) Approaches 0: $\frac{100}{n} \rightarrow 0$.

(b) Grows without bound: $\frac{100}{n} \rightarrow \infty$.

55. $\frac{k+1}{k}$ for $k > 0$: (a) rewrite as sum; (b) limit as $k \rightarrow \infty$.

(a) $\frac{k+1}{k} = 1 + \frac{1}{k}$.

(b) As $k \rightarrow \infty$, $\frac{1}{k} \rightarrow 0$, so the expression $\rightarrow 1$.

56. $\frac{t^2}{t^2+1}$: (a) values at $t = 1, 10, 100$; (b) limit; (c) does it reach it?

(a) $t = 1$: $\frac{1}{2} = 0.5$; $t = 10$: $\frac{100}{101} \approx 0.990$; $t = 100$: $\frac{10000}{10001} \approx 0.9999$.

(b) Divide top and bottom by t^2 : $\frac{1}{1+1/t^2} \rightarrow 1$.

(c) Never: the denominator $t^2 + 1$ is always strictly greater than t^2 .

57. $\frac{1}{1+n}$: (a) as $n \rightarrow 0$; (b) as $n \rightarrow \infty$.

(a) $\frac{1}{1+0} = 1$. Approaches 1.

(b) $\frac{1}{1+n} \rightarrow 0$.

58. $\frac{n}{n+1}$: (a) limit as $n \rightarrow \infty$; (b) is it always less than 1?

(a) $\frac{n}{n+1} = 1 - \frac{1}{n+1} \rightarrow 1$.

(b) Yes: $n < n+1$ always, so $\frac{n}{n+1} < 1$ for all $n > 0$.

59. Fraction $\frac{a}{b}$ with $a, b > 0$: (a) double both; (b) add 1 to both.

(a) $\frac{2a}{2b} = \frac{a}{b}$. Unchanged.

(b) **Test.** $\frac{1}{4} \rightarrow \frac{2}{5}$ (increased). $\frac{3}{4} \rightarrow \frac{4}{5}$ (increased). $\frac{3}{2} \rightarrow \frac{4}{3}$ (decreased).

(b) **Rule.** Adding 1 to both top and bottom pulls the fraction toward 1. If $\frac{a}{b} < 1$, value increases; if $\frac{a}{b} > 1$, value decreases.

60. $\left(1 + \frac{1}{n}\right)$: (a) values at $n = 1, 10, 100$; (b) limit; (c) bonus.

(a) $n = 1$: 2; $n = 10$: 1.1; $n = 100$: 1.01.

(b) $1 + \frac{1}{n} \rightarrow 1$ as $n \rightarrow \infty$.

(c) $\left(1 + \frac{1}{n}\right)^n \rightarrow e \approx 2.71828\dots$, the base of natural logarithms.

Note: The expression in (b) itself just approaches 1. It is only when raised to the n th power simultaneously that we get e . This is the fundamental limit defining e .